

EXPERIMENT: Write a Prolog Program for fruit and its color using Back Tracking.

Aim

To write a Prolog program to represent fruits and their colors and demonstrate backtracking to find multiple fruits or colors.

Objectives

- To define fruits and their corresponding colors using Prolog facts.
- To understand variables and predicates in Prolog.
- To demonstrate backtracking by finding multiple solutions.
- To execute queries for specific fruits and colors.

Algorithm

1. Start the program.
2. Define facts containing fruit names and their colors.
3. Create a predicate fruit_color/2.
4. Query the predicate with a fruit name to find its color.
5. Use the ; operator to activate backtracking and find the next solution.
6. Query with a variable to find all fruits and colors.
7. Display the results.
8. Stop the program.

CODE

```
fruit_colors = {  
    "apple": ["red", "green"],  
    "banana": ["yellow"],  
    "orange": ["orange"],  
    "grapes": ["green", "purple"],  
    "mango": ["yellow", "green"],  
    "guava": ["green"],  
    "watermelon": ["green"],
```

```

    "strawberry": ["red"]
}

def find_colors(fruit):
    if fruit in fruit_colors:
        print("\nColors of", fruit + ":")
        for color in fruit_colors[fruit]:
            print("-", color)
    else:
def find_fruits(color):
    print("\nFruits with color", color + ":")
    found = False
    for fruit, colors in fruit_colors.items():
        if color in colors:
            print("-", fruit)
            found = True
    if not found:
        print("No fruit found with this color.")
def display_all():
    print("\nAll Fruits and Colors")
    print("-" * 40)
    for fruit, colors in fruit_colors.items():
        print("=" * 45)
    print("    FRUIT AND COLOR PROGRAM")
    print("=" * 45)

while True:
    print("\n1. Find color of a fruit")

```

```
print("2. Find fruits of a color")
print("3. Display all fruits and colors")
print("4. Exit")
choice = input("\nEnter your choice: ")
if choice == "1":
    fruit = input("Enter fruit name: ").lower()
    find_colors(fruit)
elif choice == "2":
    color = input("Enter color: ").lower()
    find_fruits(color)
elif choice == "3":
    display_all()
elif choice == "4":
    print("\nProgram terminated.")
    break
else:
    print("\nInvalid choice.")
```

OUTPUT

```
*IDLE Shell 3.12.5*
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\python.exe
=====
FRUIT AND COLOR PROGRAM
=====

1. Find color of a fruit
2. Find fruits of a color
3. Display all fruits and colors
4. Exit

Enter your choice: 3

All Fruits and Colors
-----
apple : red, green
banana : yellow
orange : orange
grapes : green, purple
mango : yellow, green
guava : green
watermelon : green
strawberry : red

1. Find color of a fruit
2. Find fruits of a color
3. Display all fruits and colors
4. Exit

Enter your choice: |
```

Ln: 32 Col: 19

Result

The Prolog program was successfully implemented to represent fruits and their colors using facts. Backtracking was successfully demonstrated by retrieving multiple colors for fruits such as apple, grapes, and mango.

Conclusion

The program demonstrates how Prolog backtracking can automatically search for multiple solutions. By pressing ; after each answer, Prolog goes back to previously explored choices and finds the next matching fruit or color.